

Factory Reallocation & Shipping Optimization System

An Interactive Data-Driven Dashboard for Nassau Candy Distributor

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Abstract

This project presents an interactive Factory Reallocation & Shipping Optimization System developed for Nassau Candy Distributor. The system analyzes historical order data to monitor sales, profitability, lead time, shipping performance, operational risk, and factory performance. A Streamlit dashboard combines KPIs, dynamic key insights, interactive visualizations, optimization recommendations, and a factory reallocation scenario module. Users can filter the dashboard by region, factory, ship mode, and order year and evaluate how selected operational segments behave. The reallocation module compares a current factory with a potential target factory using historical factory-level benchmarks and reports changes in lead time and margin. The system is designed as a decision-support tool and communicates when the available historical evidence does not show a clear improvement.

Keywords

Factory Reallocation; Shipping Optimization; Data Analytics; Business Intelligence; Streamlit; Plotly; Pandas; Lead Time; Risk Analysis; Supply Chain Analytics; Decision Support.

1. Introduction

Distribution operations generate large volumes of order, shipping, factory, sales, cost, and profitability information. Static reports can make it difficult to identify operational bottlenecks and assess whether an alternative factory may provide better performance. This project converts historical operational data into an interactive analytical environment. The dashboard brings sales, profit, lead time, shipping, risk, and factory performance into one interface and connects descriptive analysis with scenario-based decision support.

2. Problem Statement

The project addresses the problem of identifying orders and factory-level situations where shipping performance or profitability may be improved through better allocation decisions. A reallocation that reduces lead time but substantially decreases margin may not be desirable. Therefore, the system evaluates multiple indicators and clearly communicates when a potential reassignment does not provide sufficient improvement according to historical benchmarks.

3. Objectives

The objectives are to analyze sales and profitability, monitor factory-wise performance, identify high-risk and slow-shipping orders, analyze average lead time, analyze shipping performance by shipping mode, compare factories using historical data, identify potential reallocation opportunities, evaluate changes in lead time and margin, and provide an interactive decision-support dashboard.

4. Data and Preparation

The project uses a Nassau Candy Distributor order dataset containing order, shipping, geographic, factory, sales, cost, and derived operational fields. Important fields include Order ID, Order Date, Ship Date, Ship Mode, Customer ID, Country/Region, City, State/Province, Lead Time (Days), Factory, Factory Latitude, Factory Longitude, Profit Margin %, Sales per Unit, Cost per Unit, Order Month, Order Quarter, Order Year, Shipping Speed, Factory Code, Division Code, Region Code, Risk Level, Recommendation Status, and Scenario ID. Excel was used for data preparation and validation, while Pandas is used in the Streamlit application for loading, filtering, grouping, and aggregation.

5. Proposed System

The system contains a data-processing layer and an interactive visualization layer. The application loads the prepared Excel dataset, applies user-selected filters, calculates KPI and factory-level summaries, and presents the results through Streamlit and Plotly. A scenario module compares a selected current factory with a potential new factory using historical factory-level benchmarks. The output can indicate improvement or return 'No Clear Improvement' when the available evidence does not support reassignment.

6. Dashboard Design

The final dashboard follows the analytical sequence: Dashboard Overview → Key Performance Indicators → Key Insights → Sales & Profit Analysis → Optimization Recommendation → Factory Reallocation Scenario → Filtered Dataset. This structure keeps summary information, detailed analysis, and decision support visually separated. The design intentionally avoids unnecessary chart density and focuses on a clear operational story.

7. Key Performance Indicators and Insights

The dashboard presents Total Orders, Total Sales, Total Profit, and Average Lead Time. The Key Insights section is placed below the KPI cards and dynamically identifies the highest-sales region, highest-gross-profit factory, factory with the highest average lead time, and counts of slow-shipping and high-risk orders. The insights respond to the active filters, making them useful for focused operational analysis.

8. Sales and Profit Analysis

The dashboard includes Sales by Region, Gross Profit by Factory, Sales by Ship Mode, Risk Level Distribution, Recommendation Status, and Average Lead Time by Factory. These visualizations help users identify regional sales patterns, factory profitability differences, shipping behavior, risk concentration, recommendation status, and lead-time differences.

9. Optimization Recommendation

The optimization section monitors high-risk orders, slow-shipping orders, and average lead time. It highlights orders that may require review for possible factory reallocation and shipping optimization. This section connects descriptive dashboard analysis with the more detailed factory scenario evaluation.

10. Factory Reallocation Scenario

The Factory Reallocation Scenario is the main decision-support component. Users compare a Current Factory with a Potential New Factory. The dashboard reports Current Orders, Lead Time Change, Margin Change, Target Factory, and an overall recommendation. The result is based on historical factory-level benchmarks rather than a guarantee of future performance. If the potential target does not demonstrate a meaningful benefit, the system displays 'No Clear Improvement'.

11. Interactive Filtering

Users can filter the dashboard by Region, Factory, Ship Mode, and Order Year. The selected filters update the major metrics, insights, visualizations, recommendation indicators, and filtered data view. This allows users to analyze specific operational segments without creating separate reports.

12. Technology Stack

Python is the primary programming language. Pandas is used for data manipulation and aggregation. Streamlit provides the interactive web dashboard framework. Plotly Express provides interactive charts. Microsoft Excel is used for data preparation and validation. Power BI was used for complementary business intelligence analysis during the project. The final Streamlit application is maintained in GitHub and deployed through Streamlit Community Cloud.

Technology	Role
Python	Core programming and dashboard logic

Pandas	Data loading, filtering, grouping and aggregation
Plotly	Interactive data visualization
Streamlit	Interactive web dashboard
Microsoft Excel	Data preparation and validation
Power BI	Business intelligence and complementary analysis

13. Implementation

The Streamlit application loads the prepared Excel dataset using Pandas and creates interactive Plotly visualizations. The application uses a wide dashboard layout and a professional visual theme. The GitHub project contains the main application file, prepared dataset, README documentation, requirements file, and Git ignore configuration. The final application is deployed as a live Streamlit dashboard.

14. Results and Current Dashboard Findings

In the demonstrated dashboard state, the system showed 8,549 total orders, total sales of \$141,783.63, total profit of \$93,442.80, and an average lead time of 1,320.84 days. Key Insights identified Pacific as the highest-sales region, Lot's O' Nuts as the highest-gross-profit factory, and Sugar Shack as the factory with the highest average lead time. The optimization section highlighted 2,051 high-risk orders and 2,051 slow-shipping orders. An example factory scenario comparing Lot's O' Nuts with Secret Factory showed a lead-time change of -0.64 days and a margin change of -17.64%, resulting in a 'No Clear Improvement' recommendation. These values represent the dashboard state demonstrated during development and depend on the active filters.

15. Discussion

The project demonstrates how interactive business intelligence can complement operational analysis. KPI and chart layers provide descriptive information, while Key Insights and the reallocation scenario move toward decision support. An important design principle is transparency: scenario results are presented as historical benchmark estimates rather than guaranteed future outcomes. The 'No Clear Improvement' condition also prevents the system from forcing a reallocation when the available evidence does not support it.

16. Limitations

The scenario evaluation is based on historical factory-level benchmarks and therefore does not guarantee future operational outcomes. Real-time inventory, transportation capacity, fuel prices, weather, carrier constraints, current factory availability, and changing demand are not fully represented. The dashboard should therefore be treated as a decision-support system rather than an autonomous allocation system.

17. Future Scope

Future improvements can include real-time shipping data integration, delivery-time prediction, cost and distance optimization, automated alerts, constraint-based factory allocation, richer predictive modeling, and integration with live operational systems. A multi-objective optimization model could simultaneously consider lead time, transportation distance, cost, margin, capacity, and service-level requirements.

18. Conclusion

The Factory Reallocation & Shipping Optimization System provides an integrated environment for analyzing order performance and evaluating potential factory reassignment decisions. By combining KPIs, dynamic insights, interactive visualizations, risk and shipping analysis, and a historical benchmark-based reallocation scenario, the project converts operational data into an accessible decision-support workflow. The final Streamlit dashboard demonstrates a practical application of Python-based analytics and business intelligence for factory and shipping optimization.

19. References

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